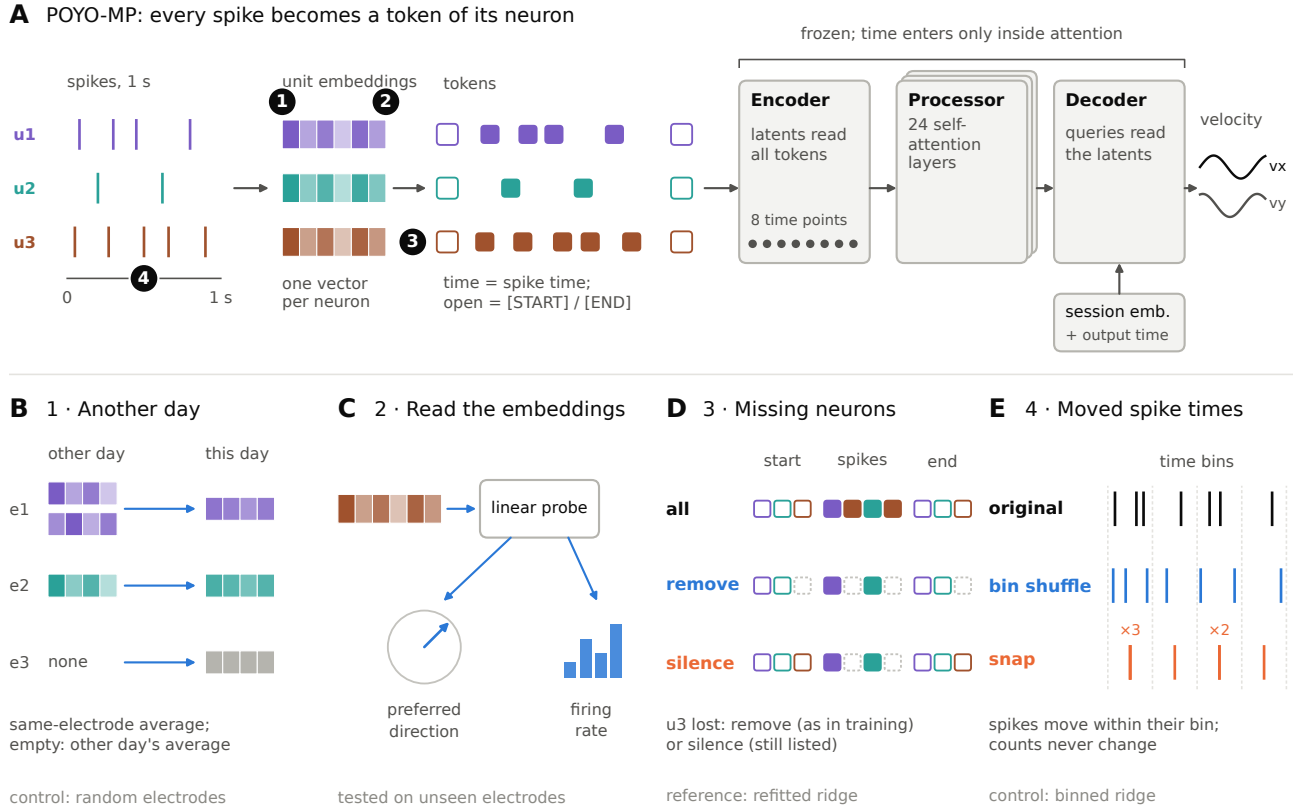
A POYO-MP: every spike becomes a token of its neuron
frozen; time enters only inside attention
spikes, 1 s
unit embeddings
tokens
u1
u2
u3
0
1 s
one vector per neuron
time = spike time; open = [START] / [END]
Encoder
latents read all tokens
8 time points
Processor
24 self-attention layers
Decoder
queries read the latents
velocity
vx
vy
session emb. + output time
B 1 · Another day
other day
this day
e1
e2
e3
none
same-electrode average; empty: other day's average
control: random electrodes
C 2 · Read the embeddings
linear probe
preferred direction
firing rate
tested on unseen electrodes
D 3 · Missing neurons
start
spikes
end
all
remove
silence
u3 lost: remove (as in training) or silence (still listed)
reference: refitted ridge
E 4 · Moved spike times
time bins
original
bin shuffle
snap
×3
×2
spikes move within their bin; counts never change
control: binned ridge